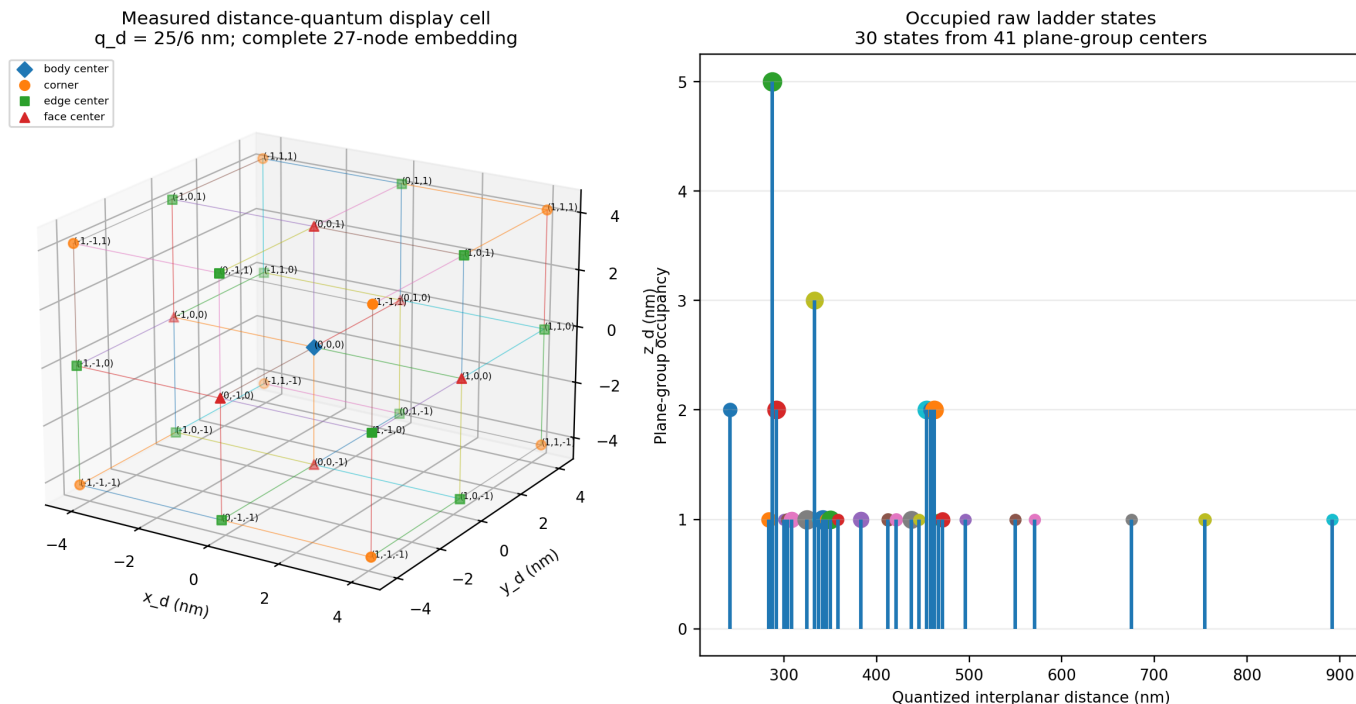


K-FIELD

SPECTROPHOTOMETER QUANTIZED INTERPLANAR DISTANCE LADDER

Raw quarter-state organization of cadence-resolved Bragg plane groups, recurrent orientation tracks, and the complete measured spacing ledger



Cover figure. Left: the complete 27-node display embedding generated directly from the measured distance quantum $q_d = 25/6$ nm. Every node is labeled by integer address and nearest-neighbor links have length q_d . Right: all 30 occupied spacing states generated by the 41 cadence-resolved plane-group centers.

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Machine-readable companion: K-Field_spectrophotometer_quantized_interplanar_distance_ladder_1785505620.json

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This paper is a standalone educational and archival technical reference. Every numerical table and figure is generated from the 1,000-frame spectrophotometer test log or from exact arithmetic transformations of the cadence-resolved phase, Bragg-solution, plane-group, recurrent-track, and histogram ledgers. No synthetic measurement series is introduced.

Central result: the reported cadence-resolved interplanar distances form an indexed ladder with principal interval $Q = 50/3$ nm and quarter-state increment $q_d = Q/4 = 25/6$ nm. The 41 reported plane-group centers occupy 30 integer q_d states. The nine recurrent plane tracks occupy nine discrete whole or quarter Q levels.

Document item	Normative value
Carrier component	$T = 54.17$ s
Sliding aperture	270 measurements = 3 nominal turns
Cadence	90 measurements = 1 nominal turn
Complete windows	9
Plane-group centers	41
Recurrent tracks	9
Principal spacing interval	$Q = 50/3$ nm = 16.6666667 nm
Quarter-state increment	$q_d = 25/6$ nm = 4.1666667 nm
Machine-readable companion	K-Field_spectrophotometer_quantized_interplanar_distance_ladder_1785505620.json

Abstract

This archival reference documents the raw quantization of interplanar distances generated by the 54.17 s component of a 1,000-frame AS7341 spectrophotometer test. The record was divided into nine overlapping 270-sample windows advanced at a 90-sample cadence. Each window supplied channel phases for the first-order two-wavelength Bragg inversion. The pair solutions were assembled into 41 cadence-resolved plane-group centers and then linked into nine recurrent orientation-distance tracks. The resulting distances are organized by the exact principal interval $Q = 50/3 \text{ nm} = 16.6666667 \text{ nm}$ and its quarter-state $q_d = Q/4 = 25/6 \text{ nm} = 4.1666667 \text{ nm}$. For each reported group distance d_g , the indexed state is $k_g = \text{round}(d_g/q_d)$, the quantized representative is $d_{\text{hat}_g} = k_g q_d$, and the raw displacement is $\epsilon_g = d_g - d_{\text{hat}_g}$. The 41 group centers occupy 30 distinct integer k states from $k = 58$ through $k = 214$. The maximum absolute displacement from the nearest occupied quarter state is 1.875644 nm ; the mean absolute displacement is 0.926310 nm . The recurrent tracks occupy 17.25Q, 18Q, 18.25Q, 19.75Q, 20Q, 20.5Q, 21Q, 25Q, and 28Q. The densely occupied ladder regions are 17Q-18.5Q, 19.5Q-21.5Q, and 24.75Q-28.25Q. Plane orientation is coupled to ladder state: the near-orthogonal 78-99 degree family occupies several distinct states, while the 7.162697 degree track occupies 28Q. The paper presents the complete raw ledgers, exact plane-family equations in the phase projection, a dimensionally correct 27-node q_d display cell, all recurrent plane cross-sections, cadence-state maps, and a machine-readable companion for direct numerical reconstruction.

Keywords: K-Field; AS7341 spectrophotometer; Bragg inversion; interplanar distance; quantized ladder; quarter state; plane orientation; cadence window; recurrent plane track.

Contents

1. Complete result statement
2. Experimental record and coordinate chain
3. Phase extraction and sliding cadence
4. Bragg plane construction
5. Raw spacing quantum and index law
6. Recurrent plane tracks
7. Complete plane-group quantization ledger
8. Occupied state distribution
9. Orientation-state coupling
10. Dimensionally correct base-cell embedding
11. Recurrent plane families and interplanar angles
12. Educational interpretation
13. Reproduction procedure

Appendices A-F: equations, phase ledger, node ledger, angle ledger, source hashes, archival record

1. Complete Result Statement

The raw interplanar-distance output forms a quantized ladder, not a continuous distribution.

$Q = 50/3 \text{ nm} = 16.6666667 \text{ nm}$
$q_d = Q/4 = 25/6 \text{ nm} = 4.1666667 \text{ nm}$
$d_k = k q_d$, with k an integer; equivalently $d_k/Q = k/4$.

The occupied states therefore consist of whole, quarter, half, and three-quarter multiples of Q .

Recurrent plane tracks

Track	Orientation (deg)	Raw d (nm)	Q level	Quantized d (nm)	Residual (nm)
R1	7.163	465.795	28	466.667	-0.872
R2	31.195	341.277	20.5	341.667	-0.390
R3	124.529	298.281	18	300.000	-1.719
R4	71.230	331.870	20	333.333	-1.464
R5	81.202	416.192	25	416.667	-0.474
R6	89.599	350.294	21	350.000	+0.294
R7	85.819	304.173	18.25	304.167	+0.006
R8	78.185	330.524	19.75	329.167	+1.357
R9	98.504	287.333	17.25	287.500	-0.167

The recurrent solutions occupy the raw states 17.25Q, 18Q, 18.25Q, 19.75Q, 20Q, 20.5Q, 21Q, 25Q, and 28Q.

Full plane-group ladder

The 41 reported plane-group centers occupy the following quarter indices:

58, 68, 69, 70, 72, 73, 74, 78, 80, 81, 82, 83, 84, 86, 92, 99, 101, 105, 107, 109, 110, 111, 112, 113, 119, 132, 137, 162, 181, 214

The most densely occupied regions are 17Q-18.5Q (283.333-308.333 nm), 19.5Q-21.5Q (325.000-358.333 nm), and 24.75Q-28.25Q (412.500-470.833 nm). Across the complete plane-group ledger, the largest displacement from the nearest quarter level is 1.875644 nm.

The plane orientations are not independent of the ladder. The near-orthogonal orientation family around 78-99 degrees occupies 17.25Q, 18.25Q, 19.75Q, 21Q, and 25Q, while the 7.162697 degree recurrent plane occupies 28Q.

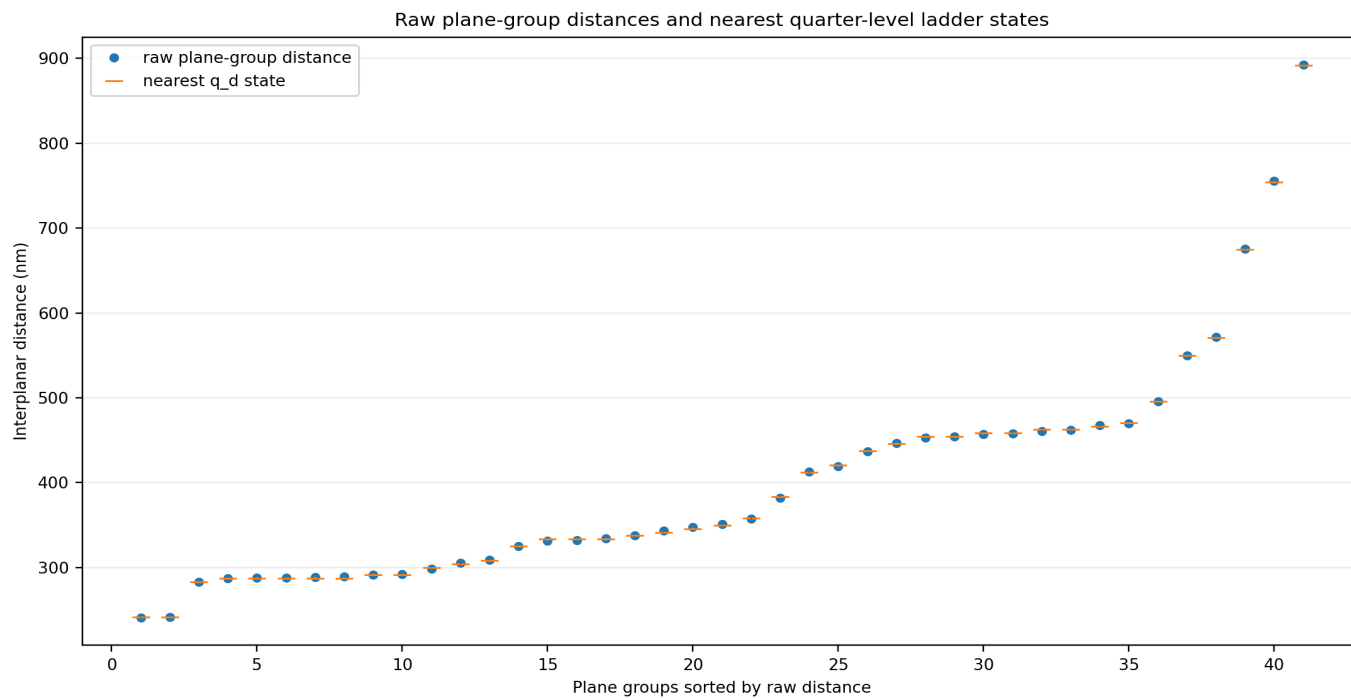


Figure 1. Complete raw plane-group distance sequence. Circles are the measured group-center distances; horizontal markers are their nearest q_d ladder representatives; vertical segments are the signed raw displacements.

2. Experimental Record and Coordinate Chain

The source test log contains 1,000 valid sequential measurements. The acquisition report records zero invalid frames, zero recovery attempts, zero all-zero transfers, and zero channel saturation events. The spectral channels used for the Bragg inversion are 415, 445, 480, 515, 555, 590, 630, 680, and nominal 910 nm NIR. Clear is retained in phase diagnostics but does not define a single wavelength for pairwise spacing inversion.

Parameter	Value
Valid frames	1,000
Carrier period	54.17 s
Window width	270 samples
Window width in nominal turns	3
Cadence	90 samples
Cadence in nominal turns	1
Complete windows	9
Wavelength-defined channels	9
Bragg pair count per window	36
Two branches per pair	2
Total branch solutions	648
Primary pair solutions	324
Plane-group centers	41
Recurrent tracks	9

The analysis chain is sequential: raw counts -> window-specific harmonic phase -> two-branch pairwise Bragg inversion -> primary branch selection -> within-window orientation/spacing groups -> cross-window recurrent tracks -> raw q_d state indexing. Every stage is retained in the machine-readable companion.

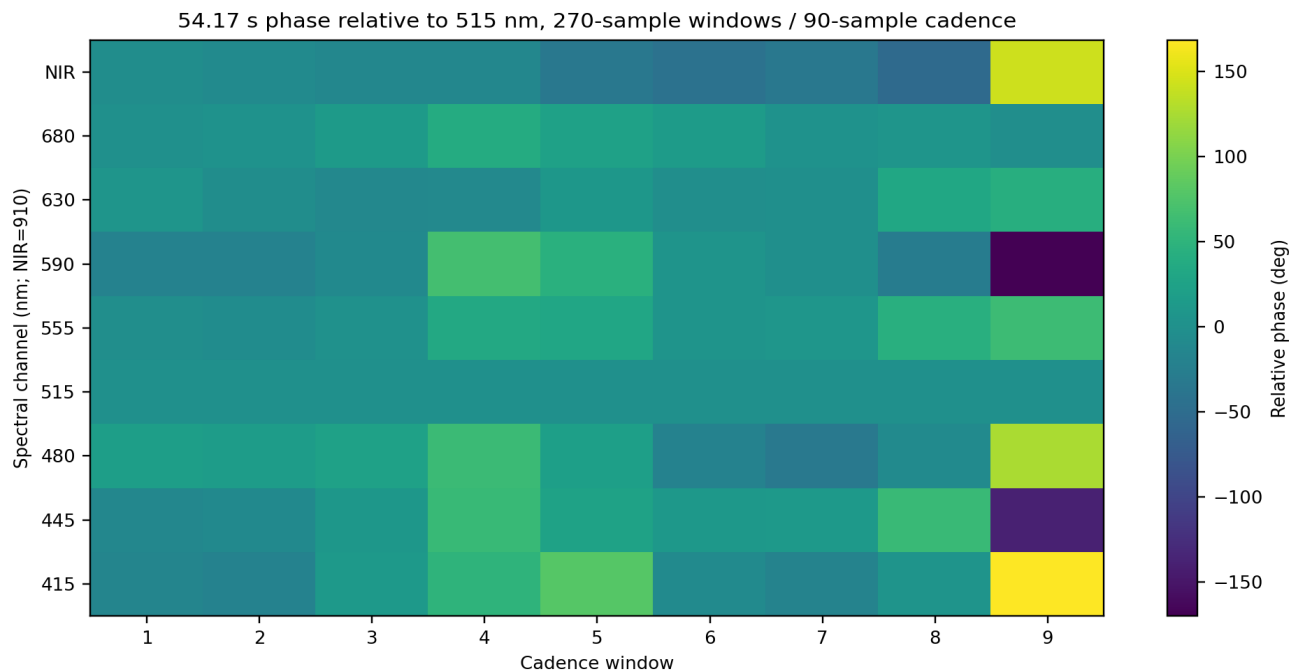


Figure 2. Measured 54.17 s channel phase relative to 515 nm in each 270-sample window. These phase rays are the angular inputs to every pairwise Bragg solution.

3. Phase Extraction and Sliding Cadence

For each channel and window, the measured count sequence $y(t)$ is decomposed into an intercept, a linear trend, and the fixed 54.17 s carrier:

$$y(t) = b_0 + b_1 (t - t_c) + A \cos(2 \pi t/T) + B \sin(2 \pi t/T)$$

The phase ray is $\phi = \text{atan2}(B, A)$, and the carrier amplitude is $\sqrt{A^2 + B^2}$. A 270-sample aperture integrates three nominal turns. Advancing the aperture by 90 samples produces one cadence step per nominal turn. Nine complete windows fit inside the 1,000-frame record.

Window-level phase-input summary

Win	Loops	Center loop	Median phase SE (deg)	Max phase SE (deg)	Median partial R2	Median amplitude
1	1-270	135.5	7.98	13.55	0.1360	2.531
2	91-360	225.5	7.79	15.50	0.1964	2.716
3	181-450	315.5	12.73	19.63	0.0642	1.766
4	271-540	405.5	24.27	38.38	0.0214	0.754
5	361-630	495.5	14.76	66.48	0.0440	1.014
6	451-720	585.5	12.33	21.43	0.0907	1.655
7	541-810	675.5	11.95	19.64	0.0655	1.615
8	631-900	765.5	35.42	87.49	0.0112	0.662
9	721-990	855.5	58.01	70.37	0.0043	0.546

The window geometry controls the temporal identity of each plane solution. Adjacent windows share 180 of 270 measurements, while their centers are separated by 90 measurements. The cadence ledger therefore records a moving three-turn view of the same carrier component.

4. Bragg Plane Construction

Each wavelength pair supplies two phase rays and two wavelengths. The first-order pairwise construction enforces a common interplanar distance for the two channels. For a surface angle s and channel phase ϕ_i , the magnitude form is:

$$d = \lambda_i / [2 |\sin(\phi_i - s)|]$$

The same-sine and opposite-sine branches are solved. The projected plane-normal axis is $n = (s + 90 \text{ deg}) \bmod 180 \text{ deg}$. The primary branch is the solution whose normal lies nearest the axial midpoint of the two phase rays. Within each cadence window, primary solutions are linked when their plane normals differ by no more than 5 degrees and their spacings differ by no more than 10 percent. Connected components with at least two pair solutions define the 41 reported plane-group centers.

Cross-window complete-link clustering under the same orientation and spacing envelope produces the nine recurrent tracks. The spacing used for a group or track is the support-weighted geometric center already present in the source ledger; the quantization step is applied afterward without changing the raw value.

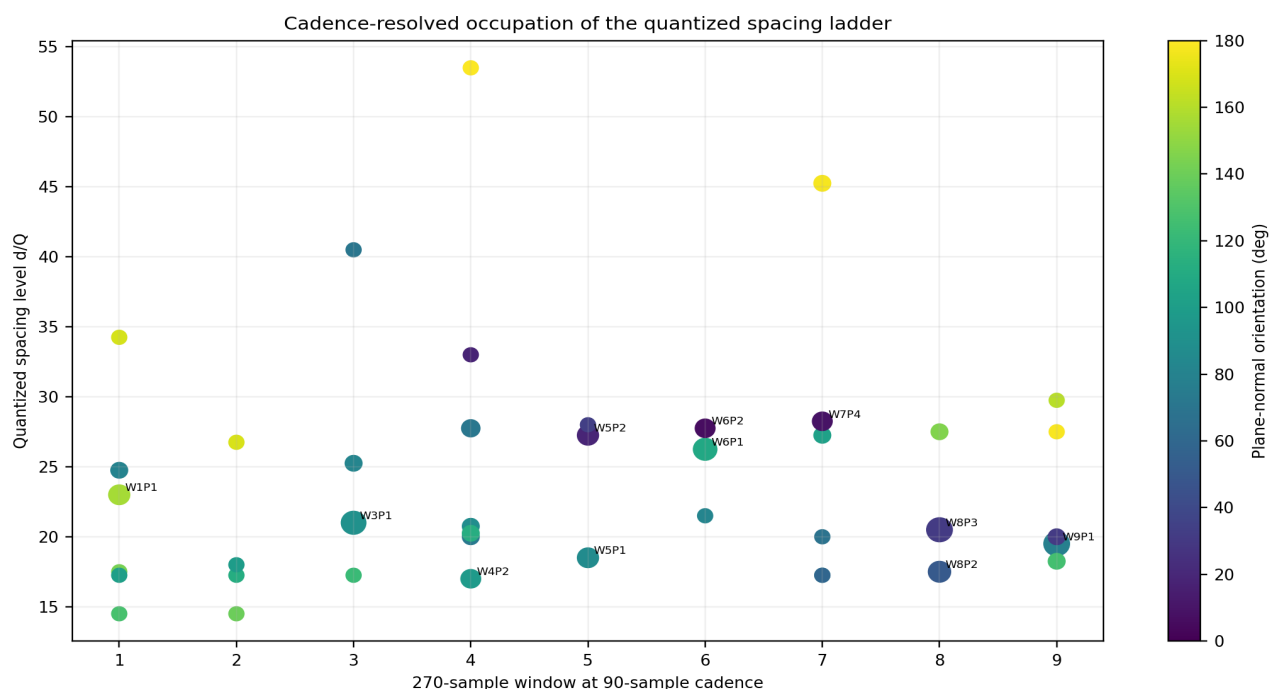


Figure 3. Cadence-resolved occupation of the quantized ladder. Marker size is pair support and marker color is plane-normal orientation. The diagram preserves both the temporal window and the discrete distance state.

5. Raw Spacing Quantum and Index Law

The principal interval is $Q = 50/3$ nm. The observed quarter subdivision is $q_d = Q/4 = 25/6$ nm. The complete indexing operation is:

$$k_g = \text{round}(d_g/q_d); \quad d_{\text{hat}}_g = k_g \, q_d; \quad \text{epsilon}_g = d_g - d_{\text{hat}}_g$$

The index k_g is an integer. The Q level is $k_g/4$. Therefore $k \bmod 4$ labels the four state classes: whole Q , quarter Q , half Q , and three-quarter Q .

Raw ladder metric	Value
Q	16.666666667 nm
q_d	4.166666667 nm
Q in SI	1.6666666667e-8 m
q_d in SI	4.1666666667e-9 m
Plane-group centers	41
Distinct occupied k states	30
Minimum occupied k	58
Maximum occupied k	214
Maximum epsilon	1.875643986 nm
Mean epsilon	0.926309594 nm
RMS epsilon	1.060546569 nm
State-class counts k mod 4	0: 9; 1: 14; 2: 12; 3: 6

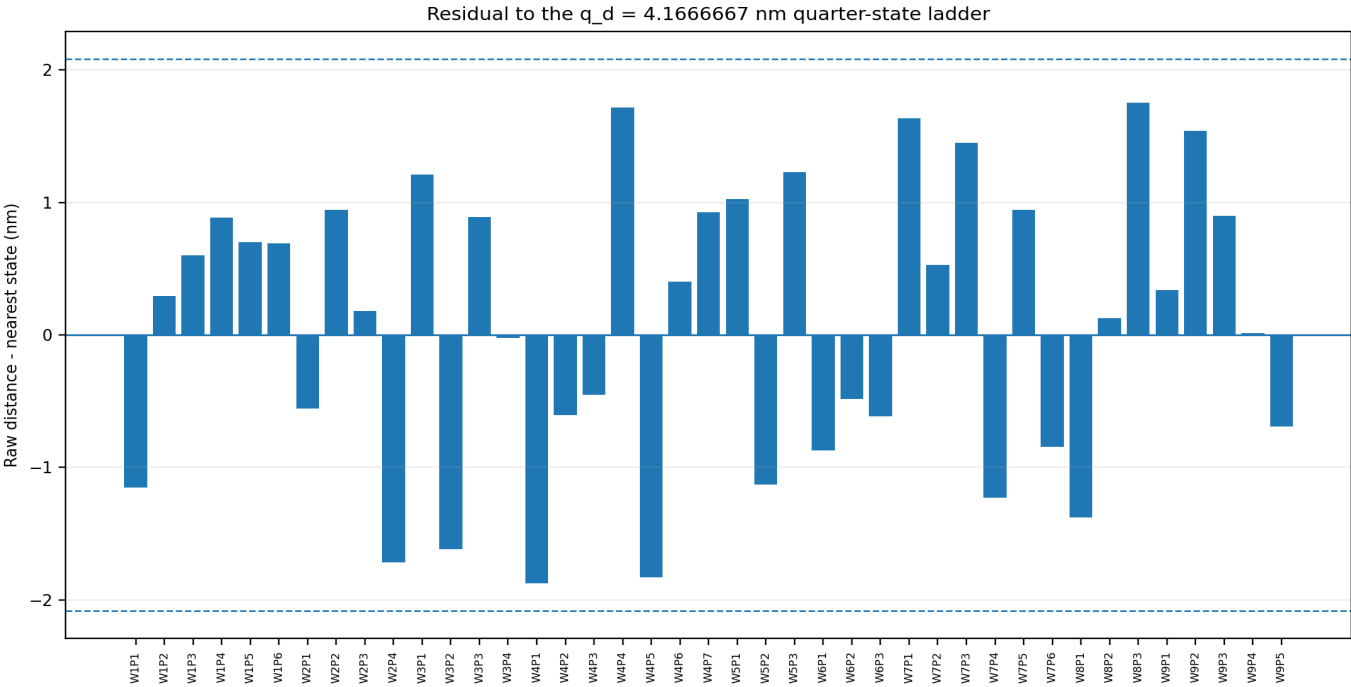


Figure 4. Signed displacement of every plane-group center from its nearest q_d state. Dashed lines are $\pm q_d/2$, the natural midpoint boundaries between adjacent indexed states.

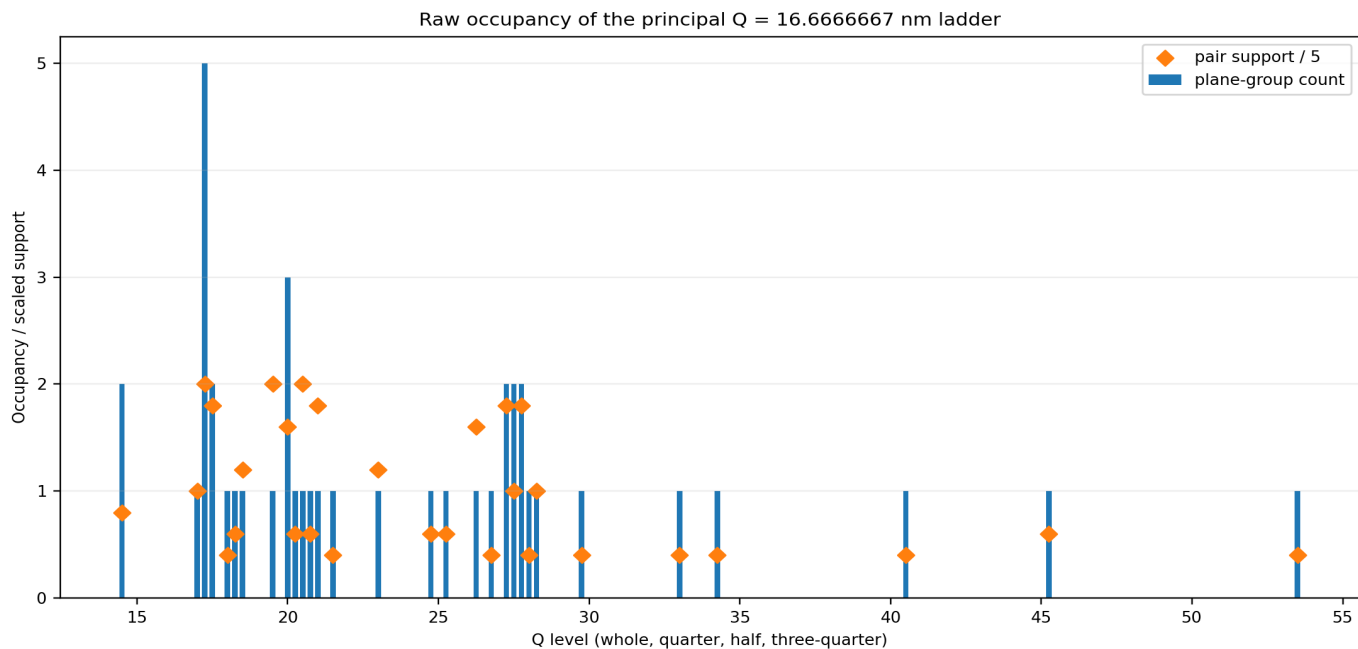


Figure 5. Occupancy of the principal Q ladder. Bars count plane groups at each Q level; diamonds show total supporting channel-pair count divided by five for a common visual scale.

6. Recurrent Plane Tracks

The recurrent-track ledger condenses repeated cadence groups into nine orientation-distance objects. Their raw distance, indexed state, quantized representative, residual, window recurrence, and total pair support are listed below.

Track	Windows	Groups	Pair support	Normal (deg)	Raw d (nm)	k	Q level	d_hat (nm)	Residual (nm)
R9	1, 2, 4	W1P5, W2P4, W4P2	9	98.503824	287.333317	69	17.25	287.500000	-0.166683
R3	3, 9	W3P4, W9P2	5	124.529258	298.281201	72	18	300.000000	-1.718799
R7	5, 7	W5P1, W7P1	8	85.818942	304.172812	73	18.25	304.166667	+0.006145
R8	6, 9	W6P3, W9P1	12	78.184811	330.523827	79	19.75	329.166667	+1.357161
R4	4, 7	W4P1, W7P6	5	71.229784	331.869612	80	20	333.333333	-1.463722
R2	8, 9	W8P3, W9P3	13	31.194545	341.276879	82	20.5	341.666667	-0.389788
R6	3, 4	W3P1, W4P4	12	89.598855	350.293683	84	21	350.000000	+0.293683
R5	1, 3	W1P6, W3P2	6	81.201814	416.192497	100	25	416.666667	-0.474170
R1	6, 7	W6P2, W7P4	10	7.162697	465.794982	112	28	466.666667	-0.871685

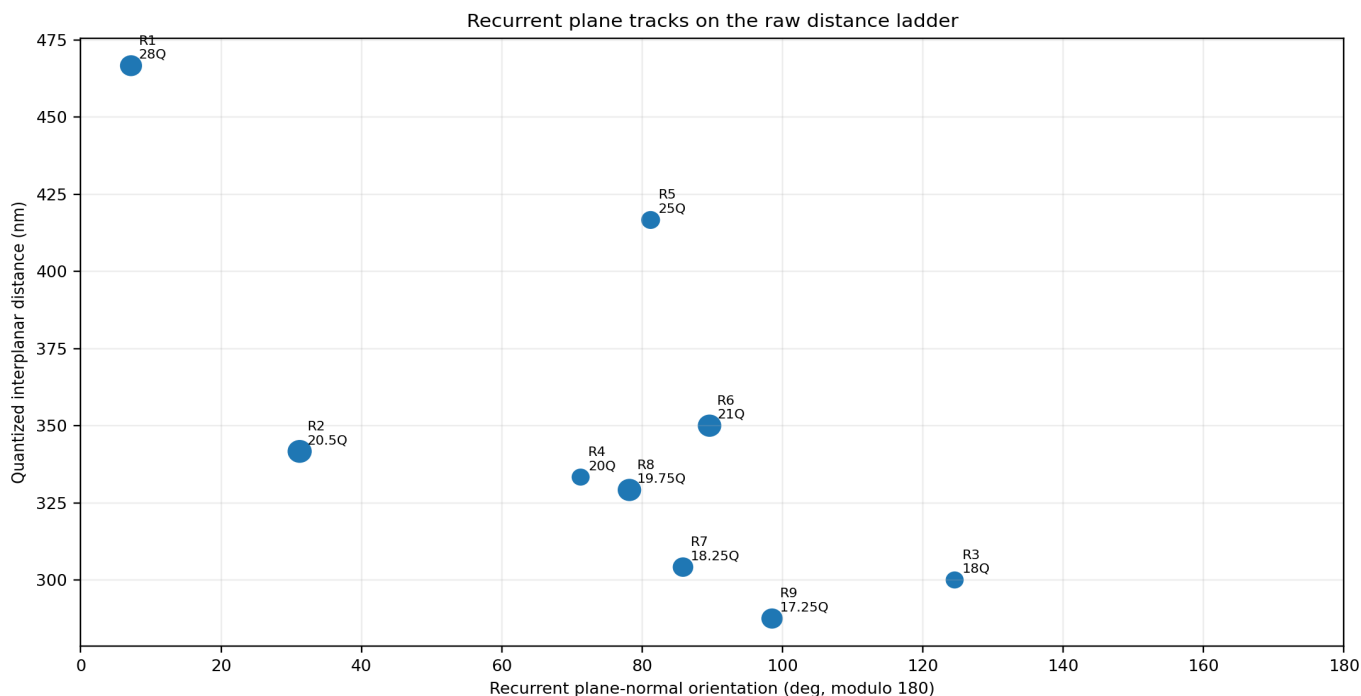


Figure 6. Nine recurrent track normals and their quantized distances. Marker size increases with total pair support. Labels report the exact Q level.

Recurrent plane-normal fan; radial coordinate is quantized Q level

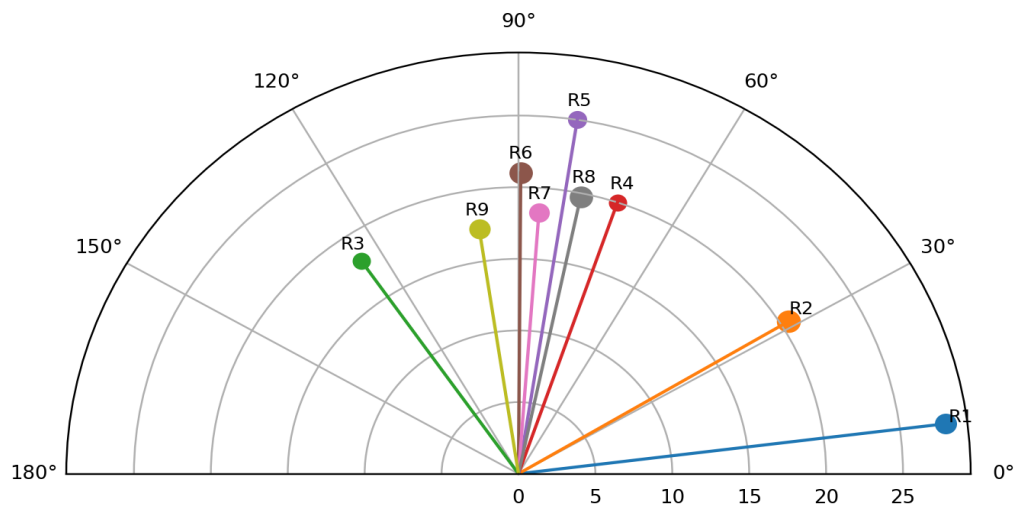


Figure 7. Recurrent plane-normal fan. Angular coordinate is the measured plane-normal orientation modulo 180 degrees; radial coordinate is the quantized Q level.

7. Complete Plane-Group Quantization Ledger

This table preserves every reported cadence-resolved plane-group center. Raw distances are not replaced; the integer state and residual are appended as derived columns.

Win	Group	Support	Normal (deg)	Raw d (nm)	k	Q level	d_hat (nm)	Residual (nm)	Pairs
1	W1P1	6	156.018218	382.179999	92	23	383.333333	-1.153335	415-445, 555-630, 445-555, 445-630, 415-630, 415-555
1	W1P2	2	127.494751	241.959760	58	14.5	241.666667	+0.293093	415-480, 445-480
1	W1P3	2	142.887062	292.266313	70	17.5	291.666667	+0.599646	415-515, 445-515
1	W1P4	2	168.299955	571.720493	137	34.25	570.833333	+0.887160	415-680, 445-680
1	W1P5	2	101.011966	288.198533	69	17.25	287.500000	+0.698533	480-555, 480-590
1	W1P6	3	80.620654	413.191276	99	24.75	412.500000	+0.691276	480-630, 630-680, 480-680
2	W2P1	2	139.301436	241.109920	58	14.5	241.666667	-0.556747	415-480, 445-480
2	W2P2	2	169.410007	446.776170	107	26.75	445.833333	+0.942837	415-630, 415-680
2	W2P3	2	111.264847	287.680097	69	17.25	287.500000	+0.180097	480-555, 480-590
2	W2P4	2	98.871992	298.282048	72	18	300.000000	-1.717952	515-590, 555-590
3	W3P1	9	90.326728	351.212701	84	21	350.000000	+1.212701	415-445, 515-590, 445-630, 445-590, 555-630, 445-515, 415-515, 415-630, 415-590
3	W3P2	3	81.782975	419.215517	101	25.25	420.833333	-1.617816	415-555, 515-630, 445-555
3	W3P3	2	70.741052	675.889591	162	40.5	675.000000	+0.889591	415-NIR, 445-NIR
3	W3P4	2	121.934095	287.479519	69	17.25	287.500000	-0.020481	480-555, 480-590
4	W4P1	3	72.960011	331.457689	80	20	333.333333	-1.875644	415-555, 415-630, 555-630
4	W4P2	5	97.354278	282.726960	68	17	283.333333	-0.606374	445-515, 515-555, 480-515, 480-555, 445-555
4	W4P3	2	17.627769	549.550602	132	33	550.000000	-0.449398	445-590, 480-590
4	W4P4	3	87.413352	347.551033	83	20.75	345.833333	+1.717700	445-630, 480-630, 630-680
4	W4P5	4	71.041940	460.670327	111	27.75	462.500000	-1.829673	445-680, 680-NIR, 480-NIR, 445-NIR
4	W4P6	3	112.137136	337.901028	81	20.25	337.500000	+0.401028	515-590, 590-680, 515-680
4	W4P7	2	178.956728	892.592963	214	53.5	891.666667	+0.926296	515-NIR, 555-680
5	W5P1	6	86.401611	309.357186	74	18.5	308.333333	+1.023853	415-515, 515-590, 415-630, 415-680, 415-590, 415-555
5	W5P2	6	18.626817	453.039276	109	27.25	454.166667	-1.127390	415-NIR, 630-NIR, 480-NIR, 480-630, 445-NIR, 445-630
5	W5P3	2	33.469916	467.896181	112	28	466.666667	+1.229515	555-NIR, 680-NIR
6	W6P1	8	108.014052	436.630398	105	26.25	437.500000	-0.869602	415-515, 590-680, 415-555, 515-680, 415-680, 415-590, 515-590, 555-680
6	W6P2	5	5.804736	462.016971	111	27.75	462.500000	-0.483029	445-630, 630-NIR, 590-NIR, 555-NIR, 445-NIR
6	W6P3	2	81.671158	357.719932	86	21.5	358.333333	-0.613402	480-630, 480-680
7	W7P1	2	84.069968	289.135186	69	17.25	287.500000	+1.635186	415-515, 415-555
7	W7P2	3	102.273489	454.693382	109	27.25	454.166667	+0.526715	415-630, 630-680, 415-680
7	W7P3	3	177.434914	755.615584	181	45.25	754.166667	+1.448918	445-555, 445-680, 555-680
7	W7P4	5	8.520658	469.603887	113	28.25	470.833333	-1.229447	445-630, 630-NIR, 590-NIR, 555-NIR, 445-NIR
7	W7P5	2	58.527808	288.445371	69	17.25	287.500000	+0.945371	480-555, 480-590
7	W7P6	2	68.632463	332.488455	80	20	333.333333	-0.844879	480-630, 480-680
8	W8P1	3	145.821385	456.955884	110	27.5	458.333333	-1.377449	415-NIR, 515-NIR, 445-NIR
8	W8P2	7	50.295361	291.791273	70	17.5	291.666667	+0.124606	445-480, 480-555, 515-555, 480-630, 515-630, 480-515, 445-515
8	W8P3	10	31.263378	343.419091	82	20.5	341.666667	+1.752425	445-555, 630-680, 590-630, 590-680, 555-590, 555-680, 555-630, 445-680, 445-630, 445-590
9	W9P1	10	77.489193	325.337931	78	19.5	325.000000	+0.337931	415-515, 515-590, 415-630, 515-630, 590-630, 555-630, 555-590, 515-555, 415-555, 415-590
9	W9P2	3	126.258051	305.706869	73	18.25	304.166667	+1.540203	445-480, 480-555, 445-555
9	W9P3	3	30.965096	334.232212	80	20	333.333333	+0.898878	445-680, 480-680, 480-590
9	W9P4	2	160.744652	495.847059	119	29.75	495.833333	+0.013725	515-NIR, 555-NIR
9	W9P5	2	178.458100	457.644538	110	27.5	458.333333	-0.688795	590-NIR, 680-NIR

8. Occupied State Distribution

Thirty distinct k states are occupied by the 41 plane-group centers. Repeated occupation at the same state may occur in different windows and at different orientations. The state ledger therefore separates the scalar spacing address from the plane-normal address.

k	Q level	d_hat (nm)	Group s	Pair support	Mean raw d	Raw interval (nm)	Mean resid	Windows	Orientations (deg)
58	14.5	241.666667	2	4	241.534840	241.110-241.960	0.424920	1, 2	127.495, 139.301
68	17	283.333333	1	5	282.726960	282.727-282.727	0.606374	4	97.354
69	17.25	287.500000	5	10	288.187741	287.480-289.135	0.695934	1, 2, 3, 7	58.528, 84.070, 101.012, 111.265, 121.934
70	17.5	291.666667	2	9	292.028793	291.791-292.266	0.362126	1, 8	50.295, 142.887
72	18	300.000000	1	2	298.282048	298.282-298.282	1.717952	2	98.872
73	18.25	304.166667	1	3	305.706869	305.707-305.707	1.540203	9	126.258
74	18.5	308.333333	1	6	309.357186	309.357-309.357	1.023853	5	86.402
78	19.5	325.000000	1	10	325.337931	325.338-325.338	0.337931	9	77.489
80	20	333.333333	3	8	332.726119	331.458-334.232	1.206467	4, 7, 9	30.965, 68.632, 72.960
81	20.25	337.500000	1	3	337.901028	337.901-337.901	0.401028	4	112.137
82	20.5	341.666667	1	10	343.419091	343.419-343.419	1.752425	8	31.263
83	20.75	345.833333	1	3	347.551033	347.551-347.551	1.717700	4	87.413
84	21	350.000000	1	9	351.212701	351.213-351.213	1.212701	3	90.327
86	21.5	358.333333	1	2	357.719932	357.720-357.720	0.613402	6	81.671
92	23	383.333333	1	6	382.179999	382.180-382.180	1.153335	1	156.018
99	24.75	412.500000	1	3	413.191276	413.191-413.191	0.691276	1	80.621
101	25.25	420.833333	1	3	419.215517	419.216-419.216	1.617816	3	81.783
105	26.25	437.500000	1	8	436.630398	436.630-436.630	0.869602	6	108.014
107	26.75	445.833333	1	2	446.776170	446.776-446.776	0.942837	2	169.410
109	27.25	454.166667	2	9	453.866329	453.039-454.693	0.827053	5, 7	18.627, 102.273
110	27.5	458.333333	2	5	457.300211	456.956-457.645	1.033122	8, 9	145.821, 178.458
111	27.75	462.500000	2	9	461.343649	460.670-462.017	1.156351	4, 6	5.805, 71.042
112	28	466.666667	1	2	467.896181	467.896-467.896	1.229515	5	33.470
113	28.25	470.833333	1	5	469.603887	469.604-469.604	1.229447	7	8.521
119	29.75	495.833333	1	2	495.847059	495.847-495.847	0.013725	9	160.745
132	33	550.000000	1	2	549.550602	549.551-549.551	0.449398	4	17.628
137	34.25	570.833333	1	2	571.720493	571.720-571.720	0.887160	1	168.300
162	40.5	675.000000	1	2	675.889591	675.890-675.890	0.889591	3	70.741
181	45.25	754.166667	1	3	755.615584	755.616-755.616	1.448918	7	177.435
214	53.5	891.666667	1	2	892.592963	892.593-892.593	0.926296	4	178.957

Dense ladder region I: 17Q-18.5Q contains 11 groups. Region II: 19.5Q-21.5Q contains 9 groups. Region III: 24.75Q-28.25Q contains 12 groups. Together these three regions contain 32 of the 41 reported plane groups.

9. Orientation-State Coupling

The spacing index is scalar, while each plane group also carries an axial orientation modulo 180 degrees. Plotting the two addresses together shows that a given orientation domain can occupy multiple distance states and that the same distance state can recur at multiple orientations.

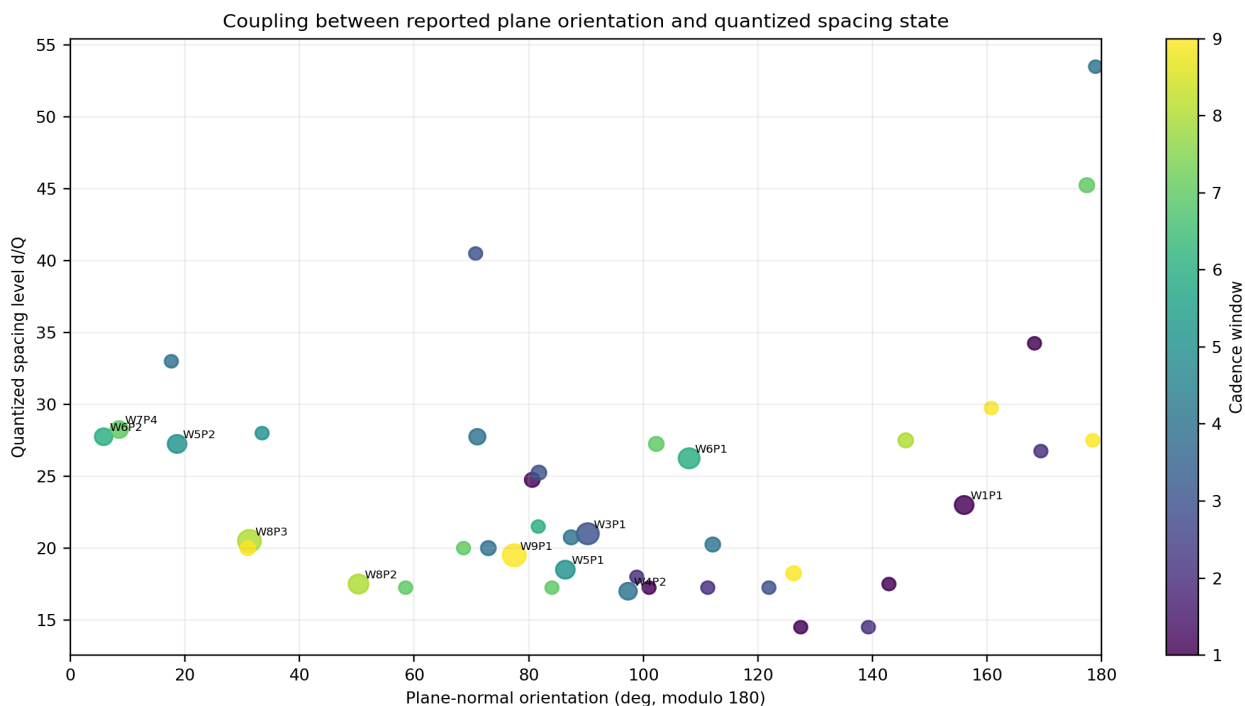


Figure 8. Plane-normal orientation versus quantized Q level for all 41 plane groups. Marker size is channel-pair support and color is cadence window.

The near-orthogonal recurrent family spans R8 at 78.184811 degrees and 19.75Q, R5 at 81.201814 degrees and 25Q, R7 at 85.818942 degrees and 18.25Q, R6 at 89.598855 degrees and 21Q, and R9 at 98.503824 degrees and 17.25Q. This is a stack of distinct spacing states within a compact orientation sector. R1 supplies the contrasting low-angle state at 7.162697 degrees and 28Q.

The state and orientation addresses therefore form a two-coordinate plane descriptor:

Plane address = (axial normal $n \bmod 180$ deg, integer spacing index k)

10. Dimensionally Correct Base-Cell Embedding

The measured quantization is one-dimensional in distance. For education and geometric bookkeeping, q_d is extended on three orthogonal display axes to form a complete $3 \times 3 \times 3$ node cell. This embedding preserves the measured length exactly: every nearest-neighbor edge has length $q_d = 4.1666667$ nm. It does not alter the measured plane normals or distances.

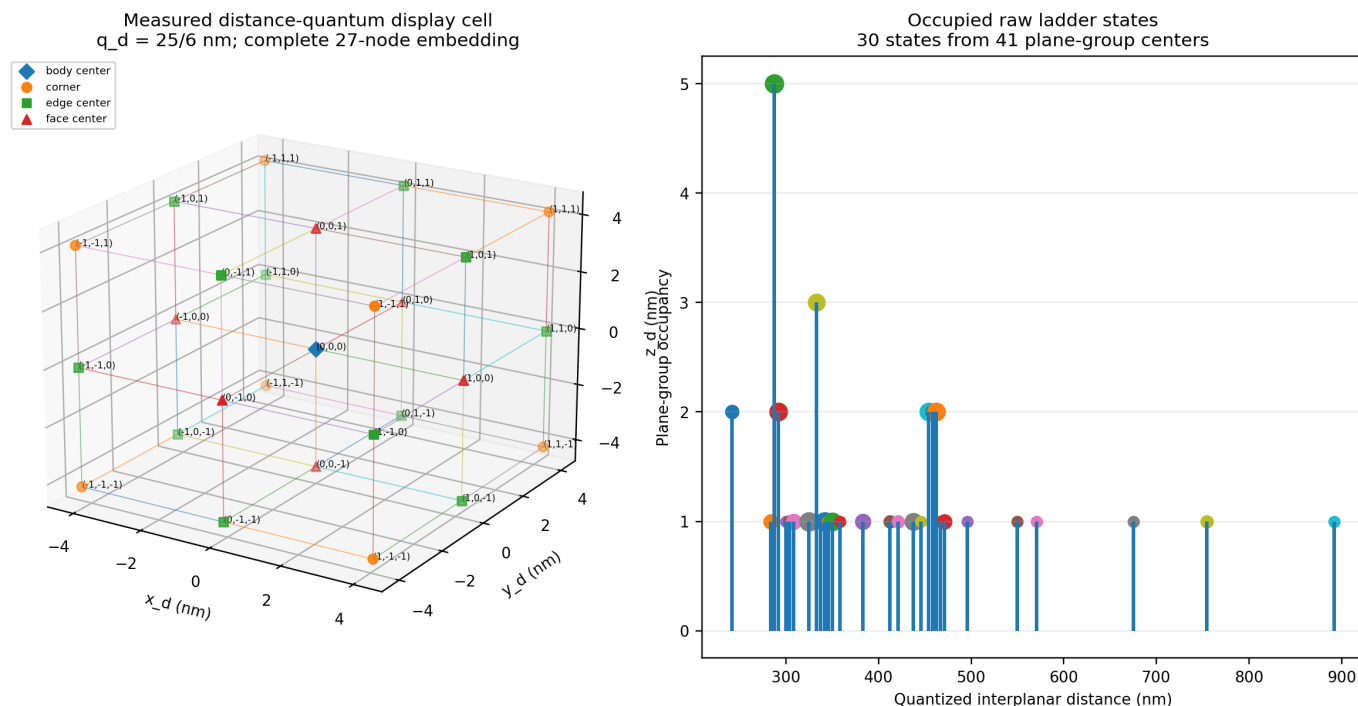


Figure 9. The left panel is the complete 27-node q_d display cell. Its physical coordinate span is $-q_d$ to $+q_d$ on each axis, so the outer-corner separation along an axis is $2q_d = 8.3333333$ nm.

Node census

Node class	Count	Coordinate rule
Corner	8	No zero address components
Edge center	12	Exactly one zero address component
Face center	6	Exactly two zero address components
Body center	1	Address (0,0,0)

The cell contains eight corners, twelve edge centers, six face centers, and one body center. The 54 nearest-neighbor links each have length q_d . All coordinates are stated in nanometres and are exact integer multiples of the measured quarter-state increment. The complete address and coordinate ledger is reproduced in Appendix C.

11. Recurrent Plane Families and Interplanar Angles

Each recurrent track defines a projected plane family with unit normal $u_i = (\cos n_i, \sin n_i, 0)$. Using the quantized spacing d_{hat_i} , the family equation is:

$$u_i \cdot r = m d_{\text{hat}_i}, \quad m \text{ integer}, \quad r = (x, y, z) \text{ in nm}$$

Dimensionally correct recurrent plane families in the measured phase projection
Lines are $m = -1, 0, +1$; arrows are plane normals

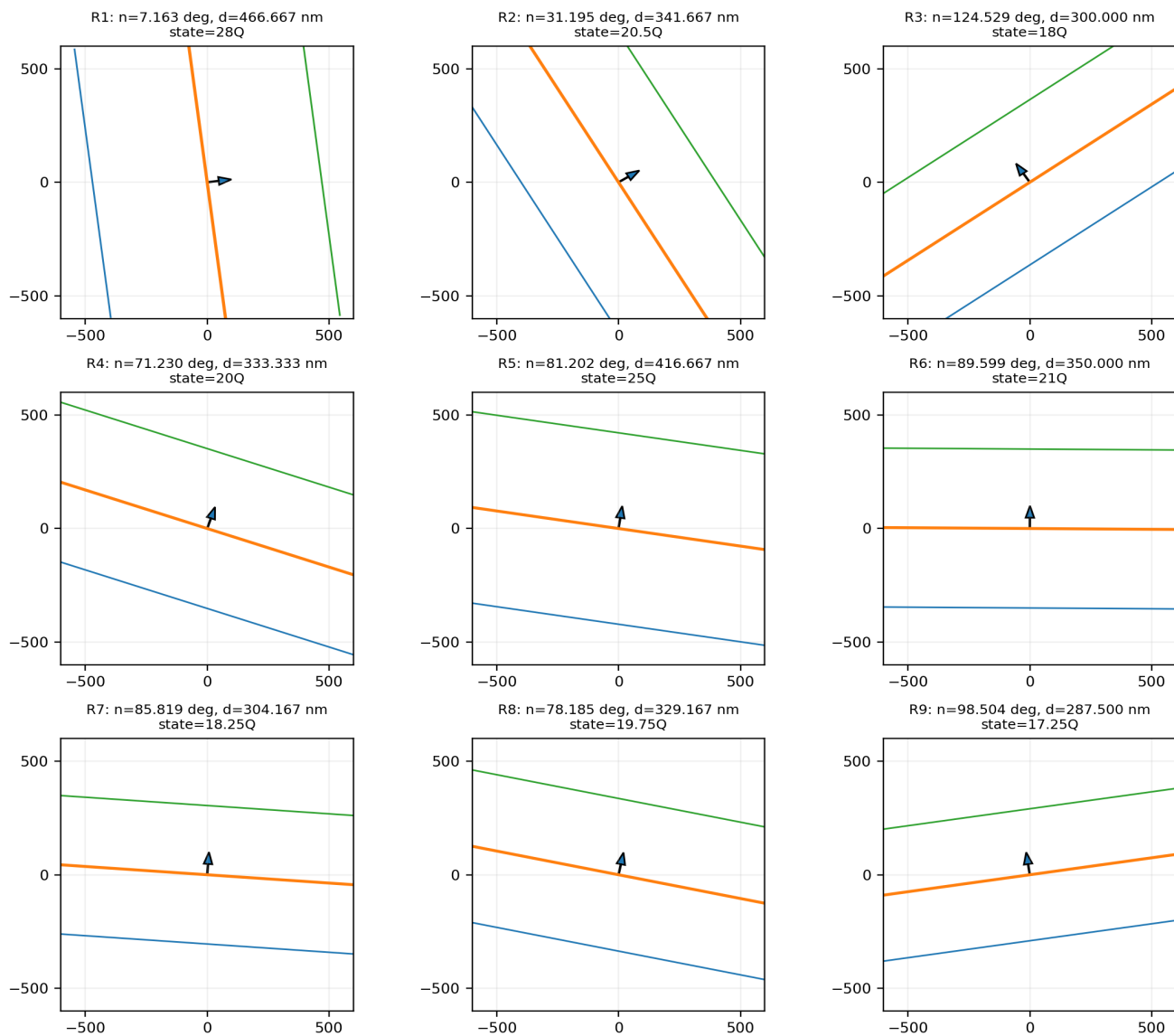


Figure 10. Dimensionally correct cross-sections of all nine recurrent plane families. Each panel shows the central plane $m=0$ and adjacent planes $m=-1$ and $m=+1$ over a ± 600 nm coordinate square. The arrow is the measured plane normal.

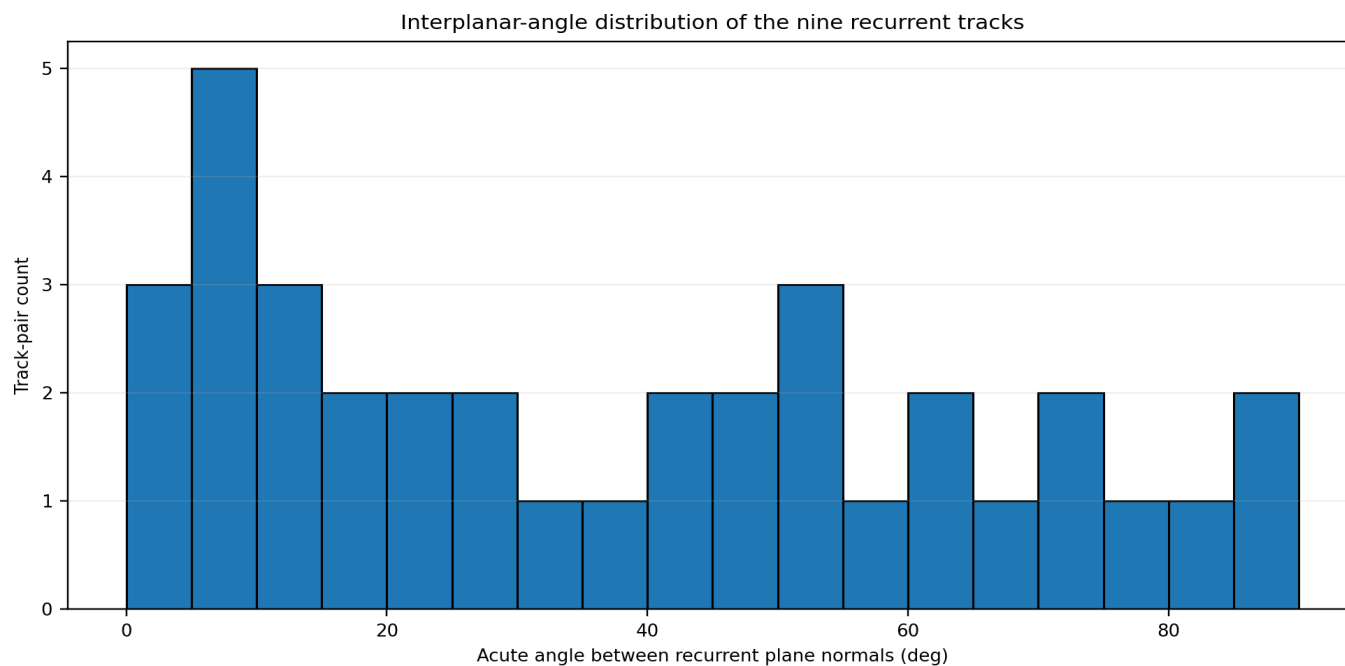


Figure 11. Distribution of the 36 acute interplanar angles generated by the nine recurrent plane normals. The smallest angle is 3.017004 degrees, the median is 36.820360 degrees, and the largest is 88.658872 degrees.

Complete recurrent-track plane equations

Track	Normal vector in phase projection	Quantized family equation
R1	(+0.992196091, +0.124687276, 0)	$0.992196091087 x + 0.124687276143 y = m * 466.666666666667 \text{ nm}; m \text{ integer}$
R2	(+0.855413581, +0.517945562, 0)	$0.855413580843 x + 0.517945562496 y = m * 341.666666666667 \text{ nm}; m \text{ integer}$
R3	(-0.566827004, +0.823836845, 0)	$-0.566827004168 x + 0.823836845101 y = m * 300.000000000000 \text{ nm}; m \text{ integer}$
R4	(+0.321773558, +0.946816655, 0)	$0.321773557598 x + 0.946816654707 y = m * 333.333333333333 \text{ nm}; m \text{ integer}$
R5	(+0.152954540, +0.988233226, 0)	$0.152954540250 x + 0.988233225821 y = m * 416.666666666667 \text{ nm}; m \text{ integer}$
R6	(+0.007001237, +0.999975491, 0)	$0.007001237333 x + 0.999975491038 y = m * 350.000000000000 \text{ nm}; m \text{ integer}$
R7	(+0.072908486, +0.997338635, 0)	$0.072908485993 x + 0.997338634903 y = m * 304.166666666667 \text{ nm}; m \text{ integer}$
R8	(+0.204755545, +0.978813142, 0)	$0.204755545212 x + 0.978813141874 y = m * 329.166666666667 \text{ nm}; m \text{ integer}$
R9	(-0.147875427, +0.989005995, 0)	$-0.147875427232 x + 0.989005994937 y = m * 287.500000000000 \text{ nm}; m \text{ integer}$

These equations preserve the observed axial phase orientation and the raw quantized distance state. They are suitable for direct plotting, ray-plane intersection, and cadence-to-cadence plane-address comparison.

12. Educational Interpretation

A continuous spacing variable can take any value in an interval. A quantized spacing ladder instead uses an integer address. Here the primitive address increment is $q_d = 25/6$ nm. Multiplying q_d by k reconstructs the allowed representative distance. Dividing k by four expresses the same state in units of Q .

For example, recurrent track R7 has raw distance 304.172812 nm. Its state index is $k = 73$. The representative is $73 q_d = 304.166667$ nm, which is 18.25 Q . The residual is +0.006145 nm. Recurrent track R6 has $k = 84$, so its representative is $84 q_d = 350$ nm = 21 Q . R1 has $k = 112$, giving 466.666667 nm = 28 Q .

The complete plane descriptor requires both distance and orientation. The integer k identifies the ladder level. The axial normal n identifies the projected plane direction. Two planes may share k while having different normals; one normal sector may contain several k values. This is why the orientation-state plot is structurally richer than either a distance histogram or an angle histogram alone.

The cadence dimension adds a third address. A plane-group record can be written as (window, n , k). Recurrent tracks compress repeated window records into persistent (n , k) objects while retaining the contributing windows and channel pairs.

The resulting framework changes the description from a list of floating-point spacings to an indexed plane-state system. The raw distance remains available as d_g , while the discrete address k and residual ϵ_g preserve the relation to the common ladder.

13. Reproduction Procedure

The complete analysis can be reproduced by the following deterministic sequence:

1. Parse the 1,000 valid source rows and construct the measurement-center time coordinate from total_us.
2. Form every complete 270-sample window beginning at sample indices 0, 90, 180, ..., 720.
3. Fit the fixed $T = 54.17$ s cosine-sine carrier with intercept and linear trend for each channel.
4. Solve both first-order two-wavelength Bragg branches for every spectral pair.
5. Select the primary branch by axial-midpoint proximity.
6. Construct within-window plane groups with the retained 5 degree / 10 percent connectivity rule.
7. Construct recurrent tracks with complete-link clustering under the same envelope.
8. Apply $Q = 50/3$ nm and $q_d = Q/4$.
9. Compute k , d_{hat} , and ϵ for every plane group and recurrent track.
10. Generate every table and figure from the resulting ledgers.

The JSON companion contains the complete plane-group and recurrent-track ledgers with the appended quantization fields, every phase fit, the complete 27-node display-cell ledger, all recurrent plane equations, all 36 recurrent interplanar angles, source-file hashes, and generation metadata.

Reproduction item	File
PDF	K-Field_spectrophotometer_quantized_interplanar_distance_ladder_1785505620.pdf
Machine-readable companion	K-Field_spectrophotometer_quantized_interplanar_distance_ladder_1785505620.json
Plane-group source	K-Field_T54p17_270x90_plane_groups_1785504180.csv
Recurrent-track source	K-Field_T54p17_270x90_recurrent_plane_tracks_1785504180.csv
Phase source	K-Field_T54p17_270x90_phase_by_window_1785504180.csv
All branch solutions	K-Field_T54p17_270x90_bragg_solutions_1785504180.csv
Source test log	spectrometer_test_log(1).txt

Appendix A. Equation Ledger

Symbol / operation	Definition
Carrier	$T = 54.17 \text{ s}$
Phase model	$y(t) = b_0 + b_1(t-t_c) + A \cos(2 \pi t/T) + B \sin(2 \pi t/T)$
Phase ray	$\phi = \text{atan2}(B, A)$
Bragg spacing	$d = \lambda / [2 \sin(\phi - s)]$
Plane normal	$n = (s + 90 \text{ deg}) \bmod 180 \text{ deg}$
Principal interval	$Q = 50/3 \text{ nm}$
Quarter state	$q_d = Q/4 = 25/6 \text{ nm}$
State index	$k = \text{round}(d/q_d)$
Quantized representative	$\hat{d} = k q_d$
Q level	$\hat{d}/Q = k/4$
Residual	$\epsilon = d - \hat{d}$
Acute axial angle	$\Delta(n_1, n_2) = [(n_1 - n_2 + 90) \bmod 180] - 90 $
Projected family	$\cos(n)x + \sin(n)y = m \hat{d}; m \text{ integer}$

Appendix B. Complete Window Phase Ledger

All phase values are the fitted 54.17 s phase rays. Relative phase is referenced to 515 nm within the same window.

Win	Channel	Wavelength (nm)	Amplitude	Phase (deg)	Rel. 515 (deg)	Partial R2	Phase SE (deg)	Loops
1	415	415	0.50367	278.841704	-15.864143	0.054372	13.5523	1-270
1	445	445	1.59707	281.909536	-12.796310	0.129477	8.3648	1-270
1	480	480	2.81449	314.536289	+19.830442	0.112758	9.0260	1-270
1	515	515	2.53097	294.705847	+0.000000	0.142869	7.7581	1-270
1	555	555	3.90421	292.175115	-2.530732	0.147199	7.6385	1-270
1	590	590	2.98510	274.965836	-19.740011	0.204794	6.4716	1-270
1	630	630	4.41383	301.105274	+6.399427	0.180547	6.7423	1-270
1	680	680	2.00751	294.837554	+0.131707	0.135998	7.9829	1-270
1	Clear		12.81263	302.018362	+7.312515	0.153070	7.4469	1-270
1	NIR	910	1.13635	291.351146	-3.354701	0.066746	11.8762	1-270
2	415	415	0.48412	287.121023	-20.451039	0.059546	15.5028	91-360
2	445	445	1.77608	298.299696	-9.272366	0.162779	8.7942	91-360
2	480	480	2.71563	324.631303	+17.059242	0.106472	10.5417	91-360
2	515	515	3.15404	307.572062	+0.000000	0.262505	6.4006	91-360
2	555	555	4.04406	301.722048	-5.850014	0.198385	7.7588	91-360
2	590	590	2.84300	287.102789	-20.469273	0.205192	7.6775	91-360
2	630	630	4.46986	303.163841	-4.408221	0.196417	7.7891	91-360
2	680	680	2.42939	310.765235	+3.193173	0.215712	7.2286	91-360
2	Clear		15.77048	315.735787	+8.163725	0.254721	6.4003	91-360
2	NIR	910	1.20943	299.361621	-8.210441	0.094170	12.0109	91-360
3	415	415	0.33827	323.590949	+12.525952	0.031656	19.6314	181-450
3	445	445	1.08592	320.814834	+9.749837	0.062328	13.6289	181-450
3	480	480	2.02714	335.232174	+24.167177	0.064211	14.1007	181-450
3	515	515	2.18179	311.064997	+0.000000	0.138666	8.4538	181-450
3	555	555	3.15030	311.793590	+0.728593	0.134355	8.6319	181-450
3	590	590	1.65919	301.947308	-9.117688	0.064683	12.5195	181-450
3	630	630	2.55952	299.703213	-11.361783	0.061960	12.7279	181-450
3	680	680	1.76616	325.514958	+14.449961	0.119350	9.7088	181-450
3	Clear		10.55238	332.091426	+21.026430	0.107622	10.5288	181-450
3	NIR	910	0.93302	298.404937	-12.660060	0.046211	14.8090	181-450
4	415	415	0.26686	303.099303	+50.173499	0.021371	24.6206	271-540
4	445	445	0.58687	311.087728	+58.161925	0.020793	24.2729	271-540
4	480	480	0.69427	313.899956	+60.974152	0.008250	38.3834	271-540
4	515	515	1.23039	252.925804	+0.000000	0.052905	16.3353	271-540
4	555	555	1.56793	287.087306	+34.161503	0.039024	18.9045	271-540
4	590	590	0.75433	320.135761	+67.209957	0.014480	28.2331	271-540
4	630	630	1.70601	242.741947	-10.183856	0.034825	19.9099	271-540
4	680	680	0.89461	291.467827	+38.542023	0.034024	20.0838	271-540
4	Clear		5.08928	329.292688	+76.366884	0.023898	21.2001	271-540
4	NIR	910	0.44479	239.389228	-13.536576	0.013454	32.0993	271-540
5	415	415	0.10627	314.220282	+79.272034	0.003089	66.4800	361-630
5	445	445	1.01354	260.256310	+25.308061	0.044031	14.7631	361-630
5	480	480	0.72778	256.300459	+21.352210	0.008508	34.1519	361-630
5	515	515	2.42492	234.948249	+0.000000	0.131548	8.3655	361-630
5	555	555	2.16590	265.274158	+30.325909	0.049115	14.0105	361-630
5	590	590	0.75099	279.514215	+44.565966	0.013531	28.0422	361-630
5	630	630	2.76715	244.631455	+9.683206	0.070337	11.5995	361-630
5	680	680	1.25870	258.797231	+23.848982	0.054133	13.2334	361-630

Win	Channel	Wavelength (nm)	Amplitude	Phase (deg)	Rel. 515 (deg)	Partial R2	Phase SE (deg)	Loops
5	Clear		4.43969	269.719952	+34.771703	0.015221	25.7971	361-630
5	NIR	910	0.63991	200.079017	-34.869232	0.025180	22.8065	361-630
6	415	415	0.34087	226.058271	-8.187458	0.030868	21.4281	451-720
6	445	445	1.35251	245.067826	+10.822098	0.100614	11.6634	451-720
6	480	480	1.33143	213.865724	-20.380004	0.035084	19.4313	451-720
6	515	515	2.80099	234.245728	+0.000000	0.209698	7.5244	451-720
6	555	555	2.53455	239.989401	+5.743673	0.085668	12.7270	451-720
6	590	590	1.78209	239.378725	+5.132996	0.090672	12.3324	451-720
6	630	630	3.41638	231.744599	-2.501130	0.136669	9.7089	451-720
6	680	680	1.65523	249.606163	+15.360435	0.124287	10.3418	451-720
6	Clear		8.39938	248.052329	+13.806601	0.070209	14.1882	451-720
6	NIR	910	0.91581	191.779242	-42.466486	0.052383	14.5860	451-720
7	415	415	0.37882	219.948902	-18.053332	0.030710	18.0195	541-810
7	445	445	1.20535	250.491419	+12.489185	0.060158	12.7368	541-810
7	480	480	1.61498	204.926615	-33.075619	0.050220	14.5186	541-810
7	515	515	2.21027	238.002234	+0.000000	0.142490	7.7691	541-810
7	555	555	2.53972	245.739955	+7.737721	0.069258	11.7076	541-810
7	590	590	1.72107	236.460205	-1.542029	0.065528	11.9513	541-810
7	630	630	3.65673	236.015843	-1.986391	0.132361	8.1016	541-810
7	680	680	1.43249	240.906870	+2.904636	0.082456	10.5876	541-810
7	Clear		8.88436	242.594531	+4.592297	0.074588	11.2000	541-810
7	NIR	910	0.68244	203.791040	-34.211194	0.028288	19.6407	541-810
8	415	415	0.24830	208.894930	+5.314934	0.016536	29.7340	631-900
8	445	445	0.16214	262.561517	+58.981521	0.001695	87.4863	631-900
8	480	480	0.66193	195.320969	-8.259026	0.009014	39.2044	631-900
8	515	515	0.95369	203.579995	+0.000000	0.039701	18.7742	631-900
8	555	555	0.90057	247.496395	+43.916400	0.011246	35.4150	631-900
8	590	590	0.58819	175.595917	-27.984078	0.008279	38.2830	631-900
8	630	630	2.05020	234.711166	+31.131171	0.054499	16.1279	631-900
8	680	680	0.81385	209.669925	+6.089930	0.038979	19.1677	631-900
8	Clear		3.85776	239.297411	+35.717415	0.018988	27.6361	631-900
8	NIR	910	0.21172	150.800113	-52.779882	0.002715	61.9202	631-900
9	415	415	0.09473	29.389444	+168.695299	0.002022	70.3331	721-990
9	445	445	0.21596	82.276624	-138.417521	0.003104	64.4059	721-990
9	480	480	0.32551	346.585710	+125.891565	0.002530	70.3669	721-990
9	515	515	0.57396	220.694145	+0.000000	0.012346	28.4290	721-990
9	555	555	0.55898	282.609819	+61.915674	0.004309	58.0054	721-990
9	590	590	0.54580	50.594375	-170.099771	0.008350	35.2519	721-990
9	630	630	1.38146	262.425965	+41.731819	0.029142	20.7527	721-990
9	680	680	0.89876	218.267038	-2.427107	0.036664	16.2543	721-990
9	Clear		3.81064	310.117430	+89.423284	0.028332	22.7701	721-990
9	NIR	910	0.22089	4.096189	+143.402044	0.002982	60.8946	721-990

Appendix C. Complete 27-Node Base-Cell Ledger

Every coordinate is an exact integer multiple of $q_d = 25/6$ nm.

Address	Class	x (nm)	y (nm)	z (nm)
(-1,-1,-1)	corner	-4.166667	-4.166667	-4.166667
(-1,-1,0)	edge center	-4.166667	-4.166667	0.000000
(-1,-1,1)	corner	-4.166667	-4.166667	4.166667
(-1,0,-1)	edge center	-4.166667	0.000000	-4.166667
(-1,0,0)	face center	-4.166667	0.000000	0.000000
(-1,0,1)	edge center	-4.166667	0.000000	4.166667
(-1,1,-1)	corner	-4.166667	4.166667	-4.166667
(-1,1,0)	edge center	-4.166667	4.166667	0.000000
(-1,1,1)	corner	-4.166667	4.166667	4.166667
(0,-1,-1)	edge center	0.000000	-4.166667	-4.166667
(0,-1,0)	face center	0.000000	-4.166667	0.000000
(0,-1,1)	edge center	0.000000	-4.166667	4.166667
(0,0,-1)	face center	0.000000	0.000000	-4.166667
(0,0,0)	body center	0.000000	0.000000	0.000000
(0,0,1)	face center	0.000000	0.000000	4.166667
(0,1,-1)	edge center	0.000000	4.166667	-4.166667
(0,1,0)	face center	0.000000	4.166667	0.000000
(0,1,1)	edge center	0.000000	4.166667	4.166667
(1,-1,-1)	corner	4.166667	-4.166667	-4.166667
(1,-1,0)	edge center	4.166667	-4.166667	0.000000
(1,-1,1)	corner	4.166667	-4.166667	4.166667
(1,0,-1)	edge center	4.166667	0.000000	-4.166667
(1,0,0)	face center	4.166667	0.000000	0.000000
(1,0,1)	edge center	4.166667	0.000000	4.166667
(1,1,-1)	corner	4.166667	4.166667	-4.166667
(1,1,0)	edge center	4.166667	4.166667	0.000000
(1,1,1)	corner	4.166667	4.166667	4.166667

Appendix D. Recurrent Interplanar-Angle Ledger

Track 1	Track 2	Normal 1 (deg)	Normal 2 (deg)	Acute angle (deg)
R5	R8	81.201814	78.184811	3.017004
R6	R7	89.598855	85.818942	3.779914
R5	R7	81.201814	85.818942	4.617127
R4	R8	71.229784	78.184811	6.955027
R7	R8	85.818942	78.184811	7.634131
R5	R6	81.201814	89.598855	8.397041
R6	R9	89.598855	98.503824	8.904969
R4	R5	71.229784	81.201814	9.972031
R6	R8	89.598855	78.184811	11.414045
R7	R9	85.818942	98.503824	12.684883
R4	R7	71.229784	85.818942	14.589158
R5	R9	81.201814	98.503824	17.302010
R4	R6	71.229784	89.598855	18.369071
R8	R9	78.184811	98.503824	20.319014
R1	R2	7.162697	31.194545	24.031848
R3	R9	124.529258	98.503824	26.025434
R4	R9	71.229784	98.503824	27.274041
R3	R6	124.529258	89.598855	34.930403
R3	R7	124.529258	85.818942	38.710316
R2	R4	31.194545	71.229784	40.035239
R3	R5	124.529258	81.201814	43.327444
R3	R8	124.529258	78.184811	46.344447
R2	R8	31.194545	78.184811	46.990266
R2	R5	31.194545	81.201814	50.007270
R3	R4	124.529258	71.229784	53.299474
R2	R7	31.194545	85.818942	54.624397
R2	R6	31.194545	89.598855	58.404311
R1	R3	7.162697	124.529258	62.633439
R1	R4	7.162697	71.229784	64.067087
R2	R9	31.194545	98.503824	67.309280
R1	R8	7.162697	78.184811	71.022114
R1	R5	7.162697	81.201814	74.039118
R1	R7	7.162697	85.818942	78.656245
R1	R6	7.162697	89.598855	82.436159
R2	R3	31.194545	124.529258	86.665286
R1	R9	7.162697	98.503824	88.658872

Appendix E. Source Integrity Record

File	SHA-256
spectrometer_test_log(1).txt	681bef37087dad25ebc71c605e0adc00f95a2b1d3b9415594a16bc109ee655fb
K-Field_T54p17_270x90_plane_groups_1785504180.csv	38ca5818818a4185343c2c9675686a314b34381139b352f9191024af7be2ad98
K-Field_T54p17_270x90_recurrent_plane_tracks_1785504180.csv	d56619ba6abaa7e0813a7434c49be84e1d3a310262882f9c57d2cc177681301f
K-Field_T54p17_270x90_phase_by_window_1785504180.csv	0d0be00f607e2b8088e703900de8016a119f09c0fbb9cbce697bf274d12e83e7
K-Field_T54p17_270x90_bragg_solutions_1785504180.csv	d0d43258865e0d04ff782594fe584a9d6ca760241d4b6a84a84d03ccec96b9cf
K-Field_T54p17_270x90_interplane_angles_1785504180.csv	d92b19160f34f91a8183c1ba78024359a18ccd26fcbc36a5c53799c9e6fd2c4d
K-Field_T54p17_270x90_histograms_1785504180.csv	ecef13bfd8797df53334ca6ecca2b6f560828fb5f095af93cb306153e2a656b3

Appendix F. Archival and Reproduction Record

Field	Value
Principal Investigator	P. Dwayne Esterline
Degrees	BS, Huntington University (1999); MBA, Indiana Wesleyan University (2008)
Organization	Krytronx, LLC.
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Generation epoch	1785505620
Generation UTC	2026-07-31T13:47:00Z
PDF	K-Field_spectrophotometer_quantized_interplanar_distance_ladder_1785505620.pdf
JSON	K-Field_spectrophotometer_quantized_interplanar_distance_ladder_1785505620.json
PDF engine	Python ReportLab
PDF fonts	Helvetica, Helvetica-Bold, Times-Roman, Courier only
Figure method	Python numerical computation and raster rendering from the exact source ledgers
Primary source	spectrometer_test_log(1).txt and K-Field_T54p17_270x90 cadence-resolved analysis ledgers
Historical format references	K-Field Spectrophotometer Optimized Geometry; K-Field Spectrophotometer Hidden Structure Assimilation; K-Field Spectrophotometer Hidden Structure Energy Analysis; K-Field Quantum Discrete Allowed States
Machine-readable scope	Complete quantization constants, raw and indexed plane groups, recurrent tracks, phase ledger, node ledger, projected plane equations, interplanar-angle ledger, source hashes, and figure manifest

END OF PROPRIETARY CONFIDENTIAL TRADE SECRET ARCHIVAL REFERENCE